Language-Conditioned QAOA for Portfolio Selection: A Bias-Controlled Study Reveals a Training-Horizon Confound

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Abstract. We flag a possible oversight in using large language models for portfolio optimization: we build a language-conditioned QAOA in which an LLM reads RavenPack news into per-asset convictions that tilt the cost Hamiltonian of a cardinality-constrained Quantum Approximate Optimization Algorithm (QAOA) portfolio optimizer. We ask, with bias controls, whether language adds out-of-sample value across two LLMs: qwen2.5-14B (Sept. 2024, released during the evaluation window) and an out-of-sample Llama 2 (cutoff Sept. 2022) on the same headlines. The qwen model beats matched random tilts by +0.76%/quarter and Llama 2 −0.23%/quarter ($p = 0.94$), which is indistinguishable from random. The benefit is noticed only for the model released into the test period, consistent with training-horizon hindsight rather than prediction—a misleading alpha. The pre-cutoff model is also smaller, a caveat we note. We contribute the pipeline, a reusable bias-control template, and a training-horizon look-ahead warning for LLM-for-finance evaluation.

Keywords: QAOA · Quantum optimization · LLM-conditioning · Look-ahead bias · Quantum finance.

1 Pipeline and Method

We focus on the use case of portfolio optimization in financial institutions. Chen et al. [1] use Black–Litterman with quantum numeric views, and Lee et al. [2] with LLM views optimized classically; we question their combination of quantum and LLM in one pipeline. We use a fixed set of Dow 30 across 2022–24, split into 378-day train and 63-day (next-quarter) test windows at each of 4 decision dates (2023-12-01 through 2024-09-04). We draw 400 random 14-of-30 subsets per date (without replacement), 1600 in total. Each asset is one qubit; we cap subsets at 14 for a tractable exact statevector simulation (2^{14} amplitudes), with the exact optimum over $\binom{14}{5} = 2002$ portfolios.

Each subset is reduced to 5 held assets under the Markowitz mean–variance objective (risk aversion $q = 2$; μ and Σ from CRSP total returns with shrinkage covariance). Holding exactly $B = 5$ of $n = 14$ makes x binary—a combinatorial choice over the 2002 feasible portfolios, and thus a candidate for a quantum optimizer.

The LLMs qwen2.5-14B (released Sept. 2024, its training plausibly spans our 2024 test window) and Llama 2 (pretraining cutoff Sept. 2022, before the test data window) are provided with RavenPack headlines (tagged, timestamped) and each decide whether to nudge μ; for each decision date the input is only headlines strictly before it, so there is no look-ahead in the news itself. The output from the LLM is a conviction $v_i \in [-1, 1]$. We inject the view Black–Litterman-style into expected returns, $\mu \to \mu + s\,v$ with $s = 0.10$ (a $\pm 10\%$ annual view), not tuned to force the optimizer to change its picks. As a control, the placebo matches the tilt's size and sparsity but randomizes which stocks (300 draws form a null), testing whether the language matters or whether any nudge of that size would do the same.

We encode the binary "pick 5 of 14" as a QUBO (mean–variance score plus a "hold exactly 5" penalty), mapping to an Ising model whose minimum energy is the best portfolio. We solve with QAOA at depth $p = 4$ ($2p = 8$ variational parameters). Our objective is not quantum advantage but whether the LLM's view changed the best portfolio at the exact optimum; the economic effect is read there, with QAOA's sampling quality reported separately.

2 Results

On the same headlines, qwen2.5-14B's changed-optimum tilts beat matched random tilts by $+0.76\%$/quarter, whereas Llama 2's do -0.23%/quarter. The edge appears only for the model whose training overlaps the test window; swapping to the out-of-sample model (market and factors held fixed) erases it—our core finding: the edge is the training horizon, not the market. The $+0.76\%$ is the language-specific premium: qwen's changed-instance effect ($+1.20\%$) minus matched random tilts ($+0.43\%$). The language input changes the exact optimum in 47–68% of instances, having an effect on selection.

Qwen's instance-level $p \approx 0$ (outside the 300-draw null), positive in 3 of 4 quarters; but the 1600 instances are subsets within only 4 decision dates, so this p is pseudo-replicated, not a time-series test. Llama 2 gives $p = 0.94$ (2 of 4 quarters), and the per-date effect is regime-dependent. QAOA at depth $p = 4$ recovers the true optimum only 7.7% of the time, a weak exact optimizer at this depth.

Limitations. Qwen (14B) is larger than Llama 2 (7B), and qwen has no published cutoff, so contamination is inferred; the study is a small-n exact simulation without transaction costs. These restrict causal attribution but do not overturn the pattern—present for the recent model, absent for the pre-cutoff one.

References

1. Chen, C., Chung, S., Goan, H.: Black-litterman portfolio optimization with noisy intermediate-scale quantum computers. arXiv:2312.00892 (2023)
2. Lee, Y., Kim, Y., Kim, J., Kim, S., Lee, Y.: LLM-enhanced Black-Litterman portfolio optimization. In: CIKM Workshop on Financial AI (2025)

Does Language Really Help a Quantum Portfolio Optimizer?

A Bias-Controlled Study and a Training-Horizon Confound in LLM-Conditioned QAOA

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The Central Question

When a backtest says "LLM news-views help a quantum portfolio optimizer," is that **signal — or an artifact?** We build a deliberately bias-controlled pipeline to catch ourselves.

Experimental Pipeline

01

Universe Selection

Dow 30, CRSP total returns — objective, fixed universe

02

LLM Conviction Generation

LLM reads RavenPack headlines time-stamped **strictly before** each decision date → per-asset ±10% view tilts

03

QAOA Cost Hamiltonian

Conviction tilt applied to QAOA cost function → hold 5 of 14 assets, evaluate held-out next quarter

04

Controls Applied

Matched random-tilt placebo; ablation vs. RavenPack sentiment; **two LLMs on identical headlines**

The Two-LLM Design

qwen2.5-14B

Training overlaps 2024 — **contaminated**

Llama 2

Sept-2022 cutoff — genuinely **out-of-sample**

Both models read the *same* RavenPack headlines. Only the training-horizon relationship to the backtest period differs.

⚠ **Key Finding — The apparent "language signal" is the LLM's hindsight:** The out-of-sample alpha collapses to indistinguishable-from-random with a genuinely out-of-sample model.

+0.76% per quarter

qwen2.5-14B Premium

Over random tilts (p≈0) — training overlaps 2024 backtest window

−0.23% per quarter

Llama 2 Premium

Over random tilts (p=0.94) — Sept-2022 cutoff, indistinguishable from random

47–68%

Optima Changed

A ±10% view changes the exact optimum in 47–68% of instances — language has real mechanism leverage on the selection.

7.7%

True Optimum Found

Shallow QAOA is a weak exact optimizer — no quantum-advantage claim is made

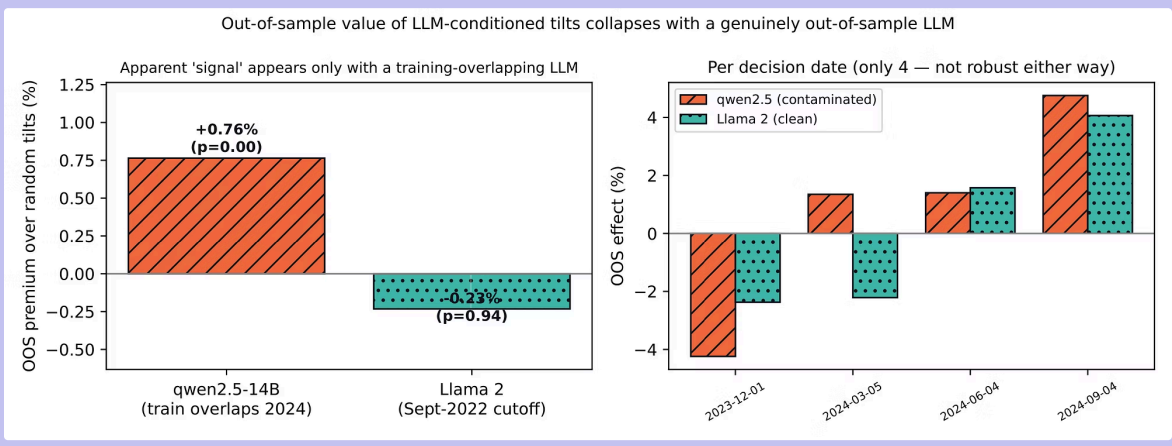


FIGURE 1 — Out-of-sample value of LLM-conditioned tilts collapses with a genuinely out-of-sample LLM.

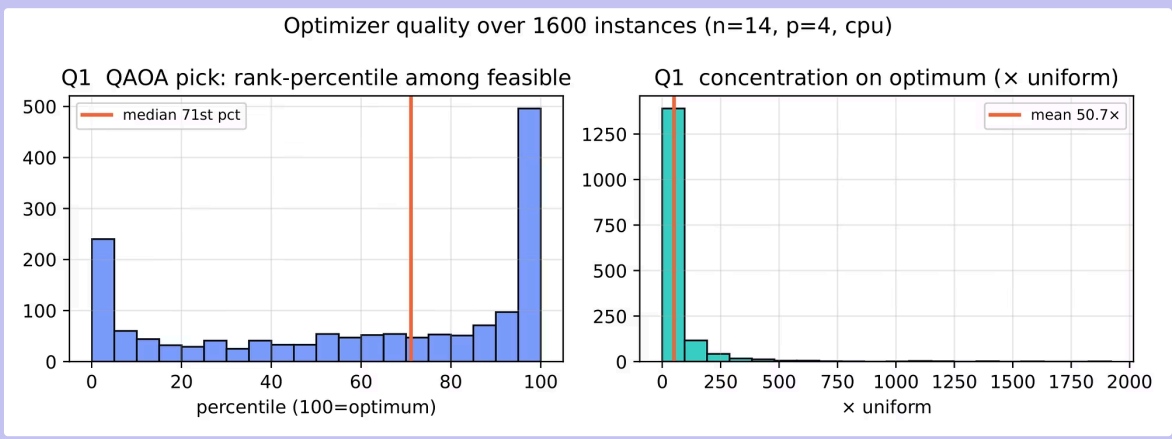


FIGURE 2 — QAOA optimizer quality over 1600 instances (n=14, p=4): rank-percentile among feasible portfolios (median 71st) and concentration on the optimum (mean 50.7× uniform).

Honest Limitations

- Only **4 decision dates** — instance-level p is pseudo-replicated
- Clean and contaminated LLMs differ in **capability** as well as cutoff date
- Small-n **exact simulation**; large-cap US equities only; no transaction costs
- None of these limitations **inflate** the central result

Contribution

→ A **reproducible bias-control template** for LLM-conditioned financial backtests

→ A **quantified warning**: date your LLM against your backtest window, or training-horizon look-ahead will manufacture alpha that isn't there

✓ **Reproducibility:** Fixed seeds, committed CRSP + conviction caches; code released upon publication.